

Responses to Comments for Paper T-ITS-23-12-3607.R1

Dear Editors & Reviewers:

We would like to express our sincere gratitude to the editor for handling our manuscript “How to Accurately Forecast Highway Traffic Speed? A Self-Supervised Spatio-Temporal Bilateral Learning Network” and to the anonymous reviewers for their insightful comments and valuable suggestions.

Based on all comments and suggestions, the authors conducted major revisions to the paper, which are summarized as follows:

- have added systematic analyses to strengthen the technical justification of key design choices, including (i) the sensitivity study for the multi-step diffusion interval parameter S (with $S = 3$ originally used in our framework), and (ii) the loss-weighting factor η in the multi-task objective (with $\eta = 0.3$ originally adopted);
- have implemented the physics-based model-driven approach in the experiment, and the code has been uploaded to our GitHub;
- have proofread the manuscript and corrected grammar/spelling issues across the full text.

Please check the detailed responses for each comment below, which are marked in blue color. All changes made to the manuscript are in yellow so that they will be easily identified. Should you have any questions, please contact us without hesitation. Once again, thank you very much for your work. We hope that the resubmitted manuscript is now suitable for publication in *IEEE Transactions on Intelligent Transportation Systems*.

Sincerely,
Dr. Guojian Zou

We thank the Editor for organizing the review for our submission. We have carefully followed the reviewers' comments and prepared a major revision in this submission. The detailed comments are addressed point-to-point as follows. We hope that the Editor and the anonymous reviewers find this revision satisfactory.

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Section #1: The entire editorial decision letter from Editor-in-Chief

Comment: 1- The Associate Editor (AE) of your manuscript, "How to Accurately Forecast Highway Traffic Speed? A Self-Supervised Spatio-Temporal Bilateral Learning Network," T-ITS-23-12-3607.R1 has recommended that we could accept your paper for publication in the Transactions on Intelligent Transportation Systems, as a REGULAR PAPER (suggested length: 10 Transactions pages) provided that you appropriately revise your paper by carefully considering the reviewers' comments listed below.

Response: We sincerely thank the Editors and the Editorial Office for the encouraging recommendation and for offering us the opportunity to revise our manuscript entitled "*How to Accurately Forecast Highway Traffic Speed? A Self-Supervised Spatio-Temporal Bilateral Learning Network*" for publication as a Regular Paper in *IEEE Transactions on Intelligent Transportation Systems*. We greatly appreciate the AE's and reviewers' time and constructive feedback.

In the revised manuscript, we have carefully addressed all reviewer comments and made corresponding revisions throughout the paper. In particular:

1. Reviewer-Requested Additional Analyses:

We added systematic analyses to strengthen the technical justification of key design choices, including (i) the sensitivity study for the multi-step diffusion interval parameter S (with $S = 3$ originally used in our framework), and (ii) the loss-weighting factor η in the multi-task objective (with $\eta = 0.3$ originally adopted). These additions provide clearer empirical evidence and training insights, as recommended.

2. Grammar, Spelling, and Readability Improvements:

We thoroughly proofread the manuscript and corrected grammar/spelling issues across the full text, while also refining phrasing for clarity and consistency.

A detailed, point-by-point response to every reviewer comment is provided in the accompanying response letter, and all corresponding changes have been incorporated into the revised manuscript. We sincerely appreciate the AE's guidance and the reviewers' helpful suggestions, which have substantially improved the quality and clarity of our work.

Comment: 2- First of all, please accept my apologies for the delay in managing your paper. Then, I include this request of mine that I am asking all authors of papers devoted to prediction in traffic.

Prediction and forecasting of traffic data have been studied in the last decades with plenty of techniques not only based on time series analysis, but more properly addressing the system dynamics. Any work about traffic forecasting must compare results with model-based approaches

that entail the physical laws behind traffic, especially highway traffic. You can consider some relevant works as:

Kurzhanskiy AA, Varaiya P (2012) Guaranteed prediction and estimation of the state of a road network. *Transp Res Part C* 21:163–180.

Roncoli C, Papageorgiou M, Papamichail I (2015) Traffic flow optimisation in presence of vehicle automation and communication systems - part I: a first-order multi-lane model for motorway traffic. *Transp Res Part C* 57:241–259.

which, among others, present very effective modelling approaches for traffic flow, traffic densities and traffic mean speeds.

Response: We sincerely thank the Editor for the careful handling of our manuscript and for the valuable and insightful comments. We fully agree with the Editor's viewpoint that traffic prediction and forecasting should be closely related to the underlying physical mechanisms of traffic flow systems. This is particularly important in highway traffic scenarios, where physics-based modeling approaches play a critical role in characterizing and predicting traffic states. In response to this comment, we have made the following reply:

1. Physical Interpretability and Proven Effectiveness:

We fully acknowledge the theoretical value and practical significance of model-based approaches in traffic research. Classical traffic flow models are grounded in conservation laws and fundamental diagrams, and they explicitly describe the evolution of key traffic variables such as flow, density, and mean speed, thereby providing clear physical meanings and strong interpretability. As demonstrated in works such as Kurzhanskiy and Varaiya (2012) and Roncoli et al. (2015), in scenarios where sensing infrastructures are sufficiently dense and system states are highly observable, these models have shown high effectiveness in traffic state estimation, prediction, and control.

2. Sparse Sensing and Limited Variables:

We would like to further clarify the objective constraints faced when applying physical-model-based approaches to real-world traffic forecasting tasks. In practical highway networks, deploying dense sensing infrastructures (such as detectors at every road segment, cross-section, toll station, ramp, and complex junction) often requires substantial economic and maintenance costs, which are difficult to afford. As a result, real traffic datasets typically contain only a limited number of traffic variables, and sensor locations are relatively sparse. In the real-world dataset used in this work, including METR-LA, PEMS-BAY, and Ningxia-YC, the available traffic information and traffic variables are relatively limited, which makes it difficult to fully support traditional model-based approaches in terms of parameter calibration and state evolution. This situation is consistent with current traffic monitoring practices. For this reason, model-based approaches are often applied in simulation environments, while a large body of existing research focuses on prediction using real-world data. Statistical methods, machine learning models, and deep learning approaches have therefore been widely adopted due to their lower requirements on data completeness and their robustness to noise and uncertainty. Our work is conducted under

realistic sensing conditions and data availability, and aims to further explore traffic speed forecasting based on deep learning methods.

3. CTM Implementation on Ningxia-YC Under Data/Boundary Limitations:

Nevertheless, we have carefully followed the Editor’s suggestion and supplemented our study with a model-based experiment tailored to the characteristics of the Ningxia-YC dataset used in this work. This dataset contains road segment speed data as well as traffic flow data measured at cross-sections, which provides a partial basis for conducting physics-based modeling experiments. Accordingly, we selected the classical Cell Transmission Model (CTM) as a representative physics-based approach and implemented it as a model-driven baseline for network-wide traffic speed prediction. It should be noted that applying CTM under the given data conditions still involves inherent limitations. From a modeling perspective, CTM predicts the evolution of traffic density, flow, and speed through traffic conservation laws, fundamental diagrams, and boundary/diverging rules. However, from a data perspective, the available flow measurements are cross-sectional rather than segment-level, and approximating cross-sectional flows as segment flows inevitably introduces errors. In addition, in multi-step forecasting scenarios, future inflows and outflows at highway entrances and exits are generally unknown, making boundary conditions difficult to specify accurately. Furthermore, in road networks containing interchanges, merges, and diverges, future traffic splitting ratios are unobservable, which further increases the uncertainty of model-based prediction. These factors jointly limit the achievable performance of physics-based models under the current data conditions. Despite these limitations, we implemented the CTM prediction process on this dataset and used it as a representative model-driven baseline. **This response and code have been uploaded to our GitHub** (<https://github.com/zouguojian/Traffic-speed-prediction/tree/main/3S-TBLN/baselines>).

The detailed experimental procedure is described as follows:

Data preprocessing: In the dataset, traffic flow is measured at cross-sections, while traffic speed is measured at the segment. To construct segment-level traffic states required by CTM, we performed the following preprocessing steps. The observed flow at a cross-section is treated as an approximation of the inflow to the segment. When a cross-section connects to multiple downstream segments, due to the lack of actual turning or diverging information, assume that the cross-sectional flow is evenly distributed among the connected downstream segments to obtain segment-level flow sequences. Given the segment-level flow and speed, traffic density is then computed based on traffic flow conservation under the LWR framework. In this way, segment-level flow, density, and speed sequences are constructed as the basic inputs for CTM inference. We emphasize that this transformation inevitably introduces approximation errors, which represent an unavoidable limitation when applying physics-based models to real-world datasets.

Parameter calibration: For CTM inference, it is necessary to first calibrate the key parameters, including the free-flow speed v_f , the congestion wave speed w , the critical density and jam density (ρ_c, ρ_j) , and the maximum capacity q_{max} . Given the presence of noise, outliers, and incomplete

state coverage in real traffic data, we adopt a statistical calibration strategy based on the training data distribution (e.g., percentile-based estimation and geometric inference), along with necessary physical feasibility constraints to ensure parameter stability and plausibility.

CTM inference: During inference, a rolling-window strategy is adopted. Starting from each initial time step in the test set, CTM performs recursive multi-step prediction for 48 future time steps, and the predicted speeds at each step are compared with ground truth observations to compute prediction errors. The major challenges in CTM inference lie in the lack of information about future boundary inflows and traffic splitting ratios at interchanges. To address this issue, we adopt the following approximations: (1) Using the training data, we compute the average entrance inflow for each time step of each week. When inference, the inflow at a future time step is assumed to be equal to the corresponding historical average, which serves as an approximate boundary condition. (2) For junctions with multiple downstream branches, due to the absence of ramp or turning ratio information, we assume the outgoing flow is evenly distributed among downstream segments. After predicting traffic density and flow at each step, the predicted density is converted to speed using the calibrated fundamental diagram parameters.

Experimental results: We conducted rolling multi-step evaluations on the test set and reported MAE, RMSE, and MAPE across different forecasting horizons. Because the parameters required for model inference are calibrated directly from the statistical distribution of the training data, the model does not require an explicit training process and therefore incurs no training time. During the inference stage, the total inference time is 21.15 s. Representative results at different prediction horizons are as follows:

- Horizon 6: MAE = 33.125, RMSE = 46.834, MAPE = 51.453%
- Horizon 12: MAE = 35.105, RMSE = 49.288, MAPE = 54.068%
- Horizon 24: MAE = 36.940, RMSE = 51.202, MAPE = 56.761%
- Horizon 48: MAE = 37.816, RMSE = 51.959, MAPE = 57.999%

In summary, model-based traffic prediction methods can provide a predictive baseline with clear physical meaning and interpretability; however, their predictive performance is constrained mainly by factors such as sparse sensor deployment, unknown boundary conditions, and limited traffic diversion information. If sufficient data becomes available in the future, we can conduct a more comprehensive study of model-based methods. We once again sincerely thank the editor for the constructive comments and look forward to your final decision.

Here, I have copied a part of them for your review. More explanations are detailed in the revised paper.

After revision (Excerpt from **Section I (Introduction), page 1**): *In the early stages, Physics-based model-driven approaches (e.g., cell transmission model, CTM) have been widely used for traffic state estimation under conservation laws constraints [1, 2]; however, as the demand for long-term prediction increased, researchers subsequently turned to time series models such as the autoregressive integrated moving average (ARIMA) and vector autoregression (VAR), which often fail to capture nonlinear correlations in traffic data.*

After revision (Excerpt from **Section V (Baseline Methods), page 8**): *CTM [1], is a physics-driven macroscopic traffic flow model that discretizes a road network into connected cells and propagates traffic density and flow over time based on conservation laws and fundamental diagrams, using sending/receiving and merge/diverge rules to enable traffic state prediction.*

After revision (Excerpt from **Section V (Experimental Results), page 9**): *The performance of CTM and ARIMA was substantially lower than that of other baseline models, highlighting the inherent challenges of long-term traffic speed prediction.*

- [1] Daganzo, C. F. (1994). The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory. *Transportation research part B: methodological*, 28(4), 269-287.
 [2] Kurzhanskiy, A. A., & Varaiya, P. (2012). Guaranteed prediction and estimation of the state of a road network. *Transportation research part C: emerging technologies*, 21(1), 163-180.

Table R1: Performance of the model-based approach

Data	Method	Horizon 6			Horizon 12			Horizon 24			Horizon 48		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Ningxia-YC	CTM	33.125	46.834	51.453%	35.105	49.288	54.068%	36.940	51.202	56.761%	37.816	51.959	57.999%
METR-LA	CTM	-	-	-	-	-	-	-	-	-	-	-	-
PEMS-BAY	CTM	-	-	-	-	-	-	-	-	-	-	-	-

Table R2: Computation cost of the model-based approach (means the model trains once on the whole training set)*

Data	Model	Parameters	Training / (iteration=100 rounds and batch size = 8)		Inference (batch = 1)	
			Time cost	GPU usage	Time cost	GPU usage
Ningxia-YC	CTM*	-	-	-	21.15 s	-
METR-LA	CTM*	-	-	-	-	-
PEMS-BAY	CTM*	-	-	-	-	-

Section #2: The entire editorial decision letter from the Associate Editor

Editor's comments:

Editor

Comments to the Author:

The paper is ready to be accepted. Please perform the analysis required by the reviewer for loss weighting and fix grammar/spelling

Response: We sincerely thank the Editor for the positive evaluation of our manuscript and for the encouraging decision to accept the paper. We highly appreciate the Editor's time and effort in handling our submission. Thank you very much for this valuable guidance. We have carefully addressed both points as requested:

(1) Analysis of Loss Weighting:

Following the reviewer's suggestion, we have conducted additional experiments to analyze the impact of the loss weighting parameter (η) in the multi-task learning objective. A sensitivity analysis was performed by varying η over a reasonable range, and the corresponding prediction performance was systematically evaluated. The results and discussion have been added to the revised manuscript to clearly demonstrate the robustness and rationality of the selected loss weight. This analysis provides further insight into how the balance between the pretext task and the main prediction task affects model performance.

(2) Grammar and Spelling Corrections:

We have carefully proofread the entire manuscript and corrected grammar, spelling, and minor language issues throughout this paper. Several sentences were also refined for improved clarity and readability, while ensuring that the technical meaning remains unchanged.

All revisions have been incorporated into the updated manuscript. We believe these changes have further strengthened the paper's quality, clarity, and completeness. Once again, we sincerely thank the Editor for the constructive comments and kind support. We hope that the revised manuscript now fully meets the journal's requirements and look forward to your final decision.

Section #3: The entire editorial decision letter from reviewer 3

Reviewer #3:

Recommendation: Accept With Minor Changes

Comments:

The encoder predicts future speeds while the decoder reconstructs historical data. This captures bidirectional temporal relationships. The model combines spatial and temporal features through fusion gates.

Comment: 1- The MSGraph-Att mechanism represents an interesting contribution to the field. It would be valuable to provide more comprehensive analysis of the design rationale. For instance, additional discussion on the selection of $S=3$ and its sensitivity could strengthen the work.

Response: We sincerely thank the reviewer for the positive assessment of the MSGraph-Att mechanism and for the constructive suggestion to further clarify its design rationale and parameter sensitivity. In response to this comment, we have made the following revisions:

(1) More Detailed Explanation of the MSGraph-Att Design:

In the revised manuscript, we have expanded the description of the working principle of the MSGraph-Att mechanism in the model introduction section. Specifically, we now provide a clearer explanation of how MSGraph-Att models multi-step traffic diffusion by treating road segments at different time steps as temporal nodes and selectively capturing the most influential spatial dependencies across multiple time intervals. This additional discussion helps to better motivate the design choices and highlight the advantage of MSGraph-Att over conventional single-step spatial attention mechanisms.

(2) Sensitivity Analysis of the Parameter S :

To further justify the selection of $S = 3$, we have conducted additional experiments on the Ningxia-YC dataset to investigate the impact of different values of S on prediction accuracy. The experimental results and corresponding analysis have been added to the experimental section. The results show that considering a small number of recent time intervals is sufficient to capture the dominant traffic diffusion patterns, while larger values of S introduce redundant information and may degrade performance due to increased noise. This analysis demonstrates that $S = 3$ achieves a good balance between modeling effectiveness and computational efficiency.

These additions strengthen the design rationale of the proposed MSGraph-Att mechanism and provide empirical evidence supporting the chosen parameter setting. We appreciate the reviewer's insightful comment, which has helped us improve the clarity and completeness of the manuscript.

Here, I have copied a part of them for your review. More explanations are detailed in the revised paper.

After revision (Excerpt from **Section IV (ST-Block Architecture), page 5**): *Previous studies often create the illusion that traffic conditions at a target road segment and a specific time step are primarily determined by its immediate neighbors or the global traffic state at the same time.*

However, this assumption implicitly ignores the physical propagation mechanism of traffic flow, in which disturbances and congestion patterns diffuse across the road network over time rather than instantaneously. In real traffic systems, the influence of upstream or spatially distant road segments typically manifests after a temporal delay, depending on factors such as travel speed, road topology, and vehicle composition.

As illustrated in Fig. 5, the traffic state of the red target road at time step t_2 is not solely determined by its local conditions at t_2 , but is strongly influenced by traffic flows originating from other road segments (i.e., blue, purple, green, gray, and orange) at an earlier time step t_1 . During the interval between t_1 and t_2 , vehicles, particularly large or slow-moving ones, gradually propagate through the network and eventually reach the target segment, leading to reduced speeds and a network-wide slowdown at t_2 . Such delayed influences cannot be adequately captured by single-step or purely spatial modeling strategies.

Motivated by this observation, we define this phenomenon as multi-step dynamic spatial correlation, which explicitly characterizes how traffic states at multiple previous time intervals jointly contribute to the current condition of a target road segment. Nevertheless, indiscriminately considering all spatio-temporal nodes may introduce redundant or irrelevant information, particularly from disconnected or weakly related road segments. Therefore, to capture multi-step dynamic spatial dependencies, we design a multi-step spatial attention mechanism (M2SFusion-Att) based on the top- k multi-head graph attention (TK-MGAT) to adaptively model spatial dependencies across multiple time steps in the road network, while mitigating unnecessary noise, as shown in Fig. 4 (c) and (d).

After revision (Excerpt from **Section V (Effect of the Number of Time Intervals)**, page 12): The number of time intervals controls the temporal range involved in the multi-step spatial modeling of the ST-Block and plays a critical role in capturing dynamic traffic diffusion patterns, as shown in Fig. R1. To examine its effect, we varied S and evaluated the corresponding prediction accuracy and computational cost. As S increased, the MAE initially decreased, indicating that incorporating multiple recent time intervals improved the modeling of delayed spatial dependencies in traffic propagation. The best performance was obtained at $S = 3$, where the MAE reached its minimum, suggesting that a small number of recent intervals was sufficient to capture the dominant diffusion effects in the road network.

When S is further increased, no additional performance gain is observed, and the MAE slightly increases. This phenomenon can be attributed to the diminishing influence of traffic states over longer temporal intervals, where additional intervals introduce redundant or noisy information into the attention aggregation process. Meanwhile, both training time and inference time increase monotonically with larger S , as the multi-step graph attention mechanism scales linearly with the number of temporal nodes. These results indicate that $S=3$ achieves a favorable balance between modeling capability and computational efficiency.



Fig. R1. Impact of time interval number (S) on MAE and time cost, including training (iteration = 100 rounds) and inference time.

Comment: 2- The choice of $\eta = 0.3$ for loss weighting might benefit from systematic exploration of alternative values. It would be helpful to discuss potential training considerations that may arise from simultaneous prediction and reconstruction objectives.

Response: We sincerely thank the reviewer for this insightful and constructive comment regarding the loss weighting strategy in the proposed multi-task learning framework. To address this concern, we have conducted additional experiments and expanded the discussion as follows:

(1) Systematic Exploration of Loss Weight η :

During the training stage, we evaluated a range of loss weights, specifically $\eta = \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$, to investigate their impact on final prediction performance. These experiments were carried out on the Ningxia-YC dataset under identical training settings. The corresponding results and analysis have been added to the experimental section of the revised manuscript. The results indicate that excessively small values of η underemphasize the reconstruction task, limiting its regularization effect, while overly large values shift the training focus away from the main prediction objective. Among the tested settings, $\eta = 0.3$ consistently achieves a favorable balance and yields the best overall prediction accuracy, thereby justifying its selection.

(2) Training Considerations of Joint Prediction and Reconstruction:

We have also expanded the discussion to clarify the training dynamics introduced by the simultaneous prediction and reconstruction objectives. In particular, the reconstruction task serves as a self-supervised regularizer, encouraging the model to learn more robust bidirectional temporal representations. However, if its contribution is weighted too heavily, the model may prioritize reconstructing historical inputs at the expense of forecasting future traffic speeds. Conversely, an insufficient reconstruction weight weakens its ability to stabilize training and mitigate error propagation. The added discussion explains how the chosen loss weighting helps balance these competing objectives and promotes stable convergence during training.

Overall, these additional experiments and analyses provide empirical and conceptual support for the chosen loss weighting strategy and offer deeper insight into the role of multi-task learning in

the proposed framework. We thank the reviewer again for this valuable suggestion, which has helped improve both the rigor and clarity of our work.

Here, I have copied a part of them for your review. More explanations are detailed in the revised paper.

After revision (Excerpt from **Section V (Effect of Loss Weight)**, pages 12 and 13): *The loss weight η controls the relative importance of the prediction objective and the reconstruction objective in the proposed bilateral learning framework. To provide a comprehensive evaluation, we conducted experiments with η ranging from 0.1 to 0.9 and reported MAE, RMSE, and MAPE, as shown in Fig. R2. The results revealed that different metrics exhibited distinct sensitivities to the choice of η . While MAE and RMSE remained relatively stable over a wide range of values, MAPE showed noticeably larger fluctuations when η deviated from a moderate setting. Among all configurations, $\eta = 0.3$ achieved the most balanced performance, yielding the lowest MAE and MAPE simultaneously, together with a competitive RMSE. This indicates that the selected loss weight improves overall prediction quality rather than favoring a single error metric.*

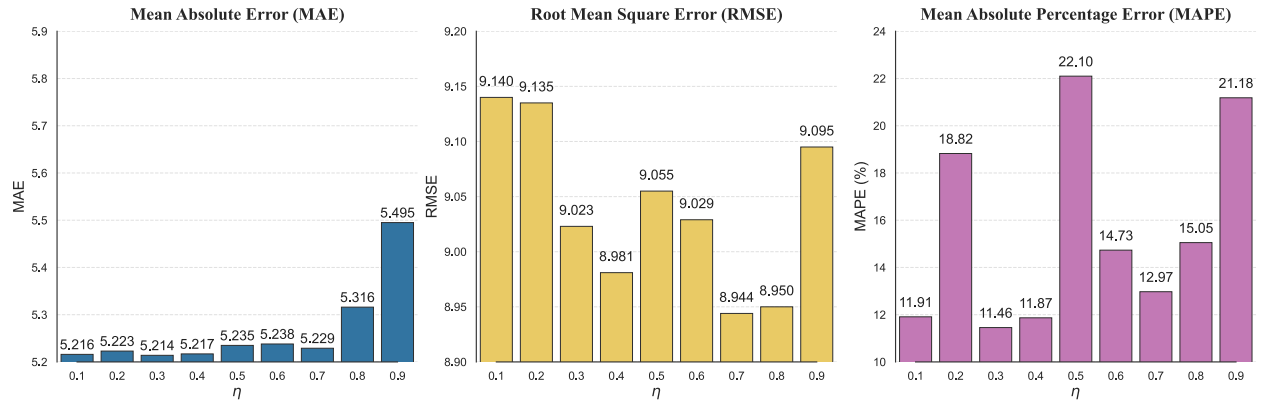


Fig. R2. Impact of loss weight (η) on MAE, RMSE, and MAPE.

Further inspection suggested that the performance degradation at larger η values was mainly driven by unstable relative errors, as reflected by the rapid increase in MAPE when $\eta \geq 0.5$. This behavior implies that overly prioritizing the reconstruction objective affects the consistency of future predictions, even though absolute errors may not increase significantly at first. From a training standpoint, an excessive reconstruction weight encourages shared representations to focus on recovering historical patterns, thereby reducing their discriminative capacity for forecasting unseen traffic states. On the other hand, very small values of η weaken the regularization effect provided by the self-supervised task. Overall, the experimental results indicate that $\eta = 0.3$ offers a suitable compromise between representation regularization and predictive accuracy, while the smooth performance variation across most settings confirms the robustness of the proposed multi-task training strategy.

Comment: 3- The 3S-TBLN framework presents a well-designed approach for traffic prediction. The technical contributions are notable and valuable. The authors might consider exploring connections with recent LLM developments. Works like “STBench: Assessing the ability of large

language models in spatio-temporal analysis”, and “UrbanGPT: Spatio-temporal large language models”.

Response: We sincerely thank the reviewer for the positive evaluation of our work and for the insightful suggestion to relate our study to recent developments in large language models (LLMs) for spatio-temporal analysis.

(1) Acknowledgement of Recent LLM-Based Spatio-Temporal Models

We fully agree that LLM-based approaches, such as UrbanGPT [1], TransGPT [2], ST-LLM [3], and related benchmarks (e.g., STBench [4]), have recently demonstrated promising performance in traffic and urban spatio-temporal prediction tasks. By fine-tuning large pre-trained models on traffic data, these methods can leverage strong prior knowledge and large parameter capacity to improve prediction accuracy on certain datasets.

(2) Differences Between LLM-Based Models and 3S-TBLN

We would like to clarify that the proposed 3S-TBLN is designed with a different focus and complementary motivation. Specifically, 3S-TBLN explicitly models traffic-specific characteristics, including the multi-step traffic diffusion process across road networks and bidirectional temporal dependencies, via the MSGraph-Att mechanism and the bilateral learning architecture. In contrast, current LLM-based approaches generally rely on implicit sequence modeling and do not explicitly encode traffic diffusion mechanisms or graph-based spatial interactions, which are critical properties of traffic systems.

(3) Practical Considerations: Model Complexity and Deployment

Another important distinction lies in practical deployment. LLM-based models typically involve a very large number of parameters, resulting in high computational, memory, and energy costs. This creates challenges for training efficiency, inference latency, and real-time deployment in intelligent transportation systems. By comparison, 3S-TBLN maintains a relatively lightweight architecture with substantially lower computational requirements, making it more suitable for real-world traffic prediction scenarios where timely response and resource efficiency are essential.

(4) Future Research Directions and Potential Integration

We agree with the reviewer that LLMs and task-specific deep learning models are not mutually exclusive. In the revised manuscript, we have added a discussion on future research directions, highlighting the potential to integrate advantages from LLMs, such as pre-training strategies, larger model capacity, and richer prior representations, into traffic-oriented frameworks like 3S-TBLN. With further advances in computational resources and data availability, such hybrid designs may further enhance model stability, robustness, and prediction accuracy.

We sincerely appreciate the reviewer’s forward-looking and insightful suggestion, which has helped us better contextualize our work within the rapidly evolving landscape of spatio-temporal modeling research.

- [1] Li, Z., Xia, L., Tang, J., Xu, Y., Shi, L., Xia, L., ... & Huang, C. (2024, August). Urbangpt: Spatio-temporal large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (pp. 5351-5362).
- [2] Wang, P., Wei, X., Hu, F., & Han, W. (2024, July). Transgpt: Multi-modal generative pre-trained transformer for transportation. In *2024 international conference on computational linguistics and Natural Language processing (CLNLP)* (pp. 96-100). IEEE.
- [3] Liu, C., Yang, S., Xu, Q., Li, Z., Long, C., Li, Z., & Zhao, R. (2024, June). Spatial-temporal large language model for traffic prediction. In *2024 25th IEEE International Conference on Mobile Data Management (MDM)* (pp. 31-40). IEEE.
- [4] Li, W., Yao, D., Zhao, R., Chen, W., Xu, Z., Luo, C., ... & Bi, J. (2025, May). Stbench: Assessing the ability of large language models in spatio-temporal analysis. In *Companion Proceedings of the ACM on Web Conference 2025* (pp. 749-752).

Here, I have copied a part of them for your review. More explanations are detailed in the revised paper.

After revision (Excerpt from **Section VI (end of Conclusion), page 15**): *In addition, with ongoing advances in computational resources and data availability, integrating LLM-inspired strategies, such as pre-training mechanisms, enhanced model capacity, and richer prior representations, into traffic-oriented frameworks like our 3S-TBLN represents a promising direction for further improving model stability, robustness, and prediction accuracy.*